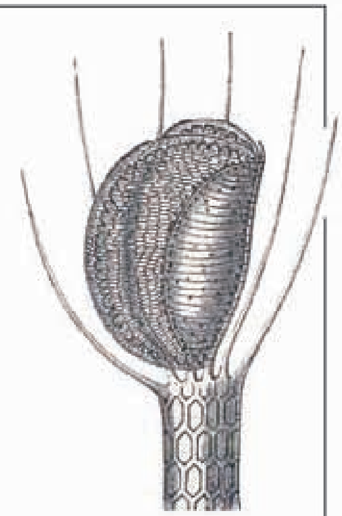
An early 19th-century painting of the Venus flytrap by Redoute, who painted the plants kept by the French Empress Josephine at Malmaison (p. 61)

GROWING A FLYTRAP

The first living specimen of the Venus flytrap arrived in England from North America in the mid-18th century. Never before had such an unusual and spectacular plant been seen live in Europe, and it aroused great curiosity among botanists. Today, Venus flytraps can be grown as pot plants. Because they come from waterlogged bogs with slightly acid soil, they must never be allowed to dry out and are best planted in peat. It is important to water them with distilled water, because tap water often contains dissolved minerals that will reduce the plant's chances of survival. Venus flytraps do produce clusters of white flowers, but this does not often happen with indoor specimens, especially if they are fed frequently with insects.

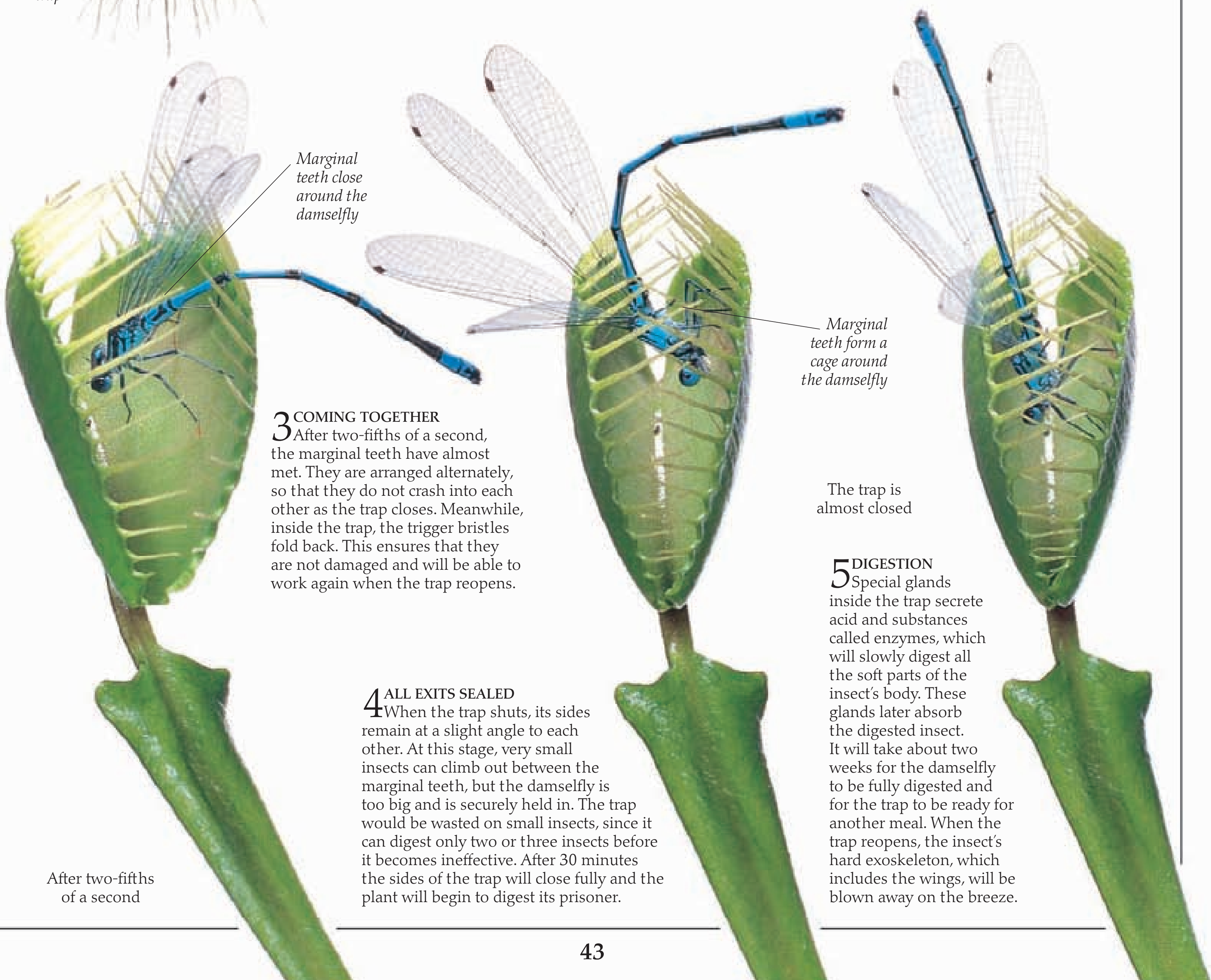
A WATERY HABITAT

Venus flytraps come from the bogs of North Carolina. Each plant grows from a small rhizome and produces several traps. Each trap can catch about three insects before it withers.



FLOATING TRAPS

This waterwheel plant is a small water plant that belongs to the same family of plant as the Venus flytrap and the sundews. Its leaves end in small traps that catch tiny water animals. They can close in one-fiftieth of a second.



3 COMING TOGETHER

After two-fifths of a second, the marginal teeth have almost met. They are arranged alternately, so that they do not crash into each other as the trap closes. Meanwhile, inside the trap, the trigger bristles fold back. This ensures that they are not damaged and will be able to work again when the trap reopens.

The trap is almost closed

5 DIGESTION

Special glands inside the trap secrete acid and substances called enzymes, which will slowly digest all the soft parts of the insect's body. These glands later absorb the digested insect. It will take about two weeks for the damselfly to be fully digested and for the trap to be ready for another meal. When the trap reopens, the insect's hard exoskeleton, which includes the wings, will be blown away on the breeze.

4 ALL EXITS SEALED

When the trap shuts, its sides remain at a slight angle to each other. At this stage, very small insects can climb out between the marginal teeth, but the damselfly is too big and is securely held in. The trap would be wasted on small insects, since it can digest only two or three insects before it becomes ineffective. After 30 minutes the sides of the trap will close fully and the plant will begin to digest its prisoner.

After two-fifths of a second